1.

Use the Windows command prompt to run the command ipconfig /all on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.

ANS:-



IPv4 address:- 192.168.0.111

Subnet Mask:- 255.255.255.0

Default Gateway:- 192.168.0.1

DNS servers:- 192.168.0.1

2.

Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to www.google.com. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.

ANS:-

Disconnected State: When your Wi-Fi or Ethernet cable is unplugged, running ping www.google.com immediately fails, returning errors such as "General failure" or "Request timed out".

- *Meaning:* The operating system has no physical data link or active route to transmit packets outside the machine, meaning local network communication is broken.

Connected State: Once reconnected, the ping command succeeds, returning reply packets with byte size, Time to Live (TTL), and round-trip time in milliseconds.

- *Meaning:* Active, bi-directional network communication has been successfully established with Google's servers over the internet gateway.

3.

Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.

ANS:-

Monitoring Tool: Windows Task Manager (Performance Tab) / Resource Monitor (resmon.exe)

Test Activity: Monitored network throughput while streaming a high-definition video on YouTube for a duration of 5 minutes.

Peak Network Usage Value: 12.5 Mbps (approximately 1.56 MB/s download throughput).

Process Consuming the Most Bandwidth: chrome.exe (Google Chrome browser process handling the active media playback tab).

4.

Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets. Hint: Use the filter bar in Wireshark and try http or https as the filter keyword.

ANS:-

- **Tool Used:** Wireshark Packet Analyzer with the application layer display filter (tls for secure HTTPS traffic).
- **Test Activity:** Captured live network traffic during 30 seconds of browsing ecommerce and social media platforms (flipkart.com / instagram.com), followed by protocol filtering.
- **3 Types of Information Visible in Filtered Packets:**
    1. **Source and Destination IP / MAC Addresses:** Identifies the local client workstation communicating with the remote web server's public IP address.
    2. **Domain Names (SNI - Server Name Indication):** In plaintext TLS *Client Hello* handshake packets, the target website domain name (flipkart.com or instagram.com) is visible before session encryption is established.
    3. **Port Numbers and Protocol State:** Displays transport layer port assignments (such as destination port 443 for secure HTTPS/TLS sessions) alongside TCP packet sequence numbers and lengths.

5.

Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.

ANS:-

Here is the troubleshooting checklist for optimizing network speed and reliability for gaming:

1. **Switch to a Wired Ethernet Connection:** Replace Wi-Fi with a direct Cat6 cable to eliminate wireless interference, packet loss, and high ping spikes.
2. **Change DNS Servers to Fast Public Resolvers:** Switch from default ISP DNS to Cloudflare (1.1.1.1) or Google (8.8.8.8) to speed up game server resolution.
3. **Enable Quality of Service (QoS):** Configure your router's QoS settings to prioritize gaming traffic over background downloads or video streaming.
4. **Update Router Firmware and Network Drivers:** Ensure your router and PC network interface cards are running the latest performance patches.
5. **Disconnect Unused Bandwidth-Heavy Devices:** Pause large background updates, cloud backups, or heavy streaming on other household devices while gaming.